

Shubham Dey

Indian Institute of Science Education and Research, Kolkata

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Education

Indian Institute of Science Education and Research, Kolkata	Aug 2022 – May 2024
M.Sc. in Physics (Astrophysics Division), CPI: 8.42/10	
Visva Bharati University	Aug 2018 – May 2021
B.Sc. in Physical Sciences (Minor in Mathematics), CPI: 8.05/10	

Publications

Orbital Stability of Moons Around the TRAPPIST-1 Planets

Dey, S. & Raymond, S. N.

The Open Journal of Astrophysics In review

[Accepted]

CLIMAX I. A Unified 1D Radiative-Convective Framework for Equilibrium and Disequilibrium Chemistry in Gas-Giant Atmospheres

Dey, S., Ghosh, P., & Majumdar, L.

The Astrophysical Journal

[In preparation]

A Fast, Parallelized, GPU-Accelerated Photochemical Model, XODIAC, with Built-in Equilibrium Chemistry and Multiple Chemical Networks for Exoplanetary Atmospheres

Ghosh, P., Mishra, S., **Dey, S.**, Das, D., Rimmer, P. B., & Majumdar, L.

The Astrophysical Journal

[Accepted]

Professional Experience

• Post-Masters Research

NISER, Bhubaneswar

Supervisor: Dr. Liton Majumdar

2024–present

Research Experience

1. Constraining the masses of exoplanets orbiting young stars through spectral feature analysis using ground-based spectroscopy. Apr 2025 – Present

Guide: Prof. Andrew Vanderburg, Harvard University; Collaborator: Dr. Saugata Barat, MIT

- Generated high-resolution transmission spectra of HIP 67522 b using `petitRADTRANS`, incorporating detailed atmospheric modeling.
- Performed injection-recovery tests with `batman` on CRIRES⁺ templates to simulate and analyze transit light curves.
- Detected atmospheric species through cross-correlation function (CCF) analysis, constructing $K_P - V_{sys}$ maps to constrain planetary signals, as part of ongoing high-resolution spectroscopy studies.

2. Stability of Orbits of Exomoons Around TRAPPIST-1 System's Planets. Oct 2024 – Present

Guide: Dr. Sean N. Raymond, Laboratoire d'Astrophysique de Bordeaux

- Studied exomoon stability using `REBOUND` N-body integrator for 10^7 orbital periods, estimated the outer stability limits of these exo-moons to be stable around the planet with and without the influence of the neighbouring planets.
- Furthermore, the results are benchmarked with [Domingos et al. \(2006\)](#) results.

3. **XODIAC: GPU-Accelerated Photochemical Model Integrating Equilibrium Chemistry and Reaction Networks for Exoplanet Atmospheres.** Oct 2024 – Jul 2025

Guide: Dr. Liton Majumdar, NISER Bhubaneswar

- Implemented the photochemistry module, calculating reaction rate coefficients through extensive literature synthesis and parameter harmonization.
- Constructed a unified reaction network from heterogeneous datasets (Arrhenius, Kooij), standardizing them to ensure numerical stability and model flexibility.
- Contributed to benchmarking XODIAC against the widely used VULCAN model, validating performance and chemical consistency.
- Collaborated with a multi-institutional development team; integrated GPU-accelerated computation to address stiffness in large kinetic networks.
- XODIAC submitted to *The Astrophysical Journal*.

4. **CLIMAX: A Unified One-Dimensional RadiativeConvective Model for Planetary Atmospheres.** Sept 2024 – Present

Guide: Dr. Liton Majumdar, NISER Bhubaneswar

- Developed a 1D radiativeconvective equilibrium model capable of simulating a wide range of atmospheres, from hot Jupiters to brown dwarfs.
- Integrated opacity tools, equilibrium and disequilibrium chemistry via FastChem and in-house Nexochem, implemented molecular and scattering cross-sections, and incorporated convective adjustment based on lapse-rate criteria to enable realistic thermal structure formation in irradiated atmospheres.
- Extended the ability of CLIMAX from equilibrium chemistry model to in-house Photochemical model, XODIAC, which I have contributed in developing, also with Photochem.
- Extending the model to self-consistently simulate climates across a wide range of planetary regimes, from hot Jupiters and brown dwarfs to terrestrial and super-Earth planets.

5. **Understanding Star-Planet Interactions Using 3D Magnetohydrodynamics Model: A Numerical Prespective** Jan 2023 – May 2024

Guides: Dr. Dibyendu Nandi & Dr. Sudip K. Garain, IISER Kolkata

- Modified PLUTO code to simulate stellar windplanetary atmosphere interactions in terms of two-fluid references and studied magnetic coupling effects.

6. **SedovTaylor Blast Wave Analysis**

Apr 2023 – Jun 2023

Guide: Dr. Sudip K. Garain, IISER Kolkata

- Simulated 3D blast waves verifying radiustime scaling laws consistent with analytic theory.

Technical Skills

Programming:	C, Python, Julia, SciLab, SQL, Matlab, Haskell (basic), Fortran, HTML
Tools:	L ^A T _E X, PowerPoint, MS Office
Computational Codes:	PLUTO, BATSRUS, Stellarium
Visualization:	VisIt, ParaView, Gnuplot, Origin
Parallel Computing:	MPI
Algebra Systems:	Mathematica
Clusters:	ASIAA-Kawas, ASIAA-xl, CESSI, TACC
GUI Development:	Tkinter, PyQt, pyfiglet, rich

Workshops and Presentations and Posters

- **Introductory Summer School in Astronomy and Astrophysics (ISSAA)** MayJul 2021
Hosted by IUCAA, India
- **Exoplanet Origins, Atmospheres and Interiors** Oct 2024
Hosted by NISER, India
- **CLIMAX: A Unified 1D Planetary Climate Model from Gas Giants to Terrestrial Exoplanets in the JWST Era** Mar 2026
EXPLORING EXOPLANETS: THE JOURNEY TO DISCOVER NEW FRONTIERS AND HABITABLE WORLDS (International conference).
Hosted by ICTS, India

Teaching Experience

Electricity and Magnetism (PH 1201) Non-certified
IISER Kolkata

Assisted with course organization and evaluation support.

Fundamentals of Astronomy and Space-sciences Certified
NISER Bhubaneshwar

Assisted teaching and evaluation for Fundamentals of Astronomy and Space Sciences, including mentoring students in research writing and term paper presentations.

Achievements

CSIR-NET (National Eligibility Test)	All India Rank 159
Graduate Aptitude Test in Engineering (GATE)	All India Rank 1679
Joint Entrance Screening Test (JEST)	All India Rank 170
Joint Admission Test for Masters (JAM)	All India Rank 137
National Graduate Physics Examination (NGPE)	Top 1%

Awards

- **DST-INSPIRE** fellow (2018 – 2021), a scholarship awarded to top 1% student in India.
- **Integrated MS-PhD Fellow** (2022 – 2024), awarded by IISER Kolkata to selected students admitted to Integrated PhD program.
- Awarded **CSIR-NET Fellowship** (Dec 2023) for qualifying

Leadership and Extracurricular Activities

- **Team Member** of Science Club and Music Club Core Committees, IISER-K.
- **Core member** of Journal club, CESSI.
- **Core member** of Outreach team, CESSI.
- Completed **Visharad** in Hindustani Classical Music and Tabla from Lucknow University, India.
- Completed **B.Mus.** and **M.Mus.** in Rabindra Sangeet from Chandigarh University, India.

===== References =====

References

Dr. Liton Majumdar

Reader-F, NISER Bhubaneswar

Email: liton@niser.ac.in

Prof. Andrew Vanderburg

Professor, Harvard University

Email: avanderburg@cfa.harvard.edu

Dr. Sean N. Raymond

CNRS Directeur de Recherche, LAB Bordeaux

Email: sean.raymond@u-bordeaux.fr